GGG-W-340B
October 17, 1974
SUPERSEDING
Fed. Spec. GGG-W-340a
March 6, 1963

FEDERAL SPECIFICATION

WIRE TWISTER PLIER (WITH SIDE CUTTER)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers wire twister pliers which are used primarily for twisting safety wires on critical equipment on aircraft, automotive equipment, and other similar applications.

1.1.1 Federal specification coverage. Federal specifications do not include all types and classes of the commodity indicated by the titles of the specifications, but cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Type and sizes. The pliers shall be of one type only and of the following sizes as specified (see 6.1):

9-inch nominal length.

12-inch nominal length.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

QQ-W-423 - Wire, Steel, Corrosion-Resisting.
PPP-P-40 - Packaging and Packing of Hand Tools.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Construction. Wire twister pliers shall be of lap joint, plier-type construction and shall incorporate a device which, when actuated, will cause the tool to revolve about its longitudinal axis. Each half of the basic plier shall include a jaw, handle, and joint section which shall be forged integral. The halves of the plier shall be permanently joined together with a through rivet which shall act as the pivot or fulcrum point of the tool. A locking device shall be included which, when engaged, shall hold the handles in a closed position in such a manner as to cause the jaws to firmly and securely grip and hold strands of lockwire.

3.2 Material. Materials used in the construction of wire twister pliers shall be as specified and shall be of uniform quality, free of injurious defects. Strength and other physical properties of all material shall be adequate for the specific purpose and shall meet or exceed the applicable requirements, including test requirements, listed herein.

3.3 Surface finish. Surfaces of wire twister pliers shall be free from pits, nodules, burrs, excess oxide scale, or other surface defects that impede operation of the tool or are uncomfortable to the hand.

3.4 Coatings. Unless otherwise specified (see 6.1), the surfaces of wire twisters, except ground and polished surfaces, shall be coated with a chemically produced oxide coating and dry oil treated.

3.5 Identification marking. Each wire twister plier shall be stamped or otherwise permanently marked in a plain and legible manner with the manufacturer's name or with a trademark of such known character that the source of manufacture may be readily determined.

3.6 Handles. Handles shall be forged from alloy steel, hardened and drawn to a Rockwell "C" value of 36-50, and shaped so as to provide a safe, comfortable handgrip.

3.7 Jaws. Jaws shall be forged from alloy steel, hardened and drawn to a Rockwell "C" value of 36-50, and shall be provided with an integral wire cutter adjacent to the joint section. The gripping surfaces of the jaws shall be of a design that will improve the grip of the jaws and securely hold the lock wire being twisted. When the gripping surfaces are of a scored or serrated design, there shall be no damage to the wire being twisted due to deep or sharp edged scores or serrations. In addition to the heat treatment above, the gripping portion of the jaws and wire cutter shall be heat treated to a hardness of Rockwell C 56-64. A wire clearance opening shall be provided between the two jaws and located between the gripping and cutting areas of the jaws. This opening shall be of a size and design which will permit the user to swiftly and easily deflect double strands of wire out of the sides of the jaws and away from the cutting edges when performing continuous (multiple) lock wiring operations. The nose design shall be as shown in figures 1 and 2 and blend with the radius of the sides of the plier in accordance with good commercial practice. The maximum permissible opening at the top of the jaws, when closed, shall be 0.010 inch. Jaw configuration known commercially as "duck billed" and "needle nosed" will not be acceptable. The wire cutter shall meet the tests as specified in 4.3.3.

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3.8 Joint rivet. The joint rivet shall be made of medium carbon steel (grade 1035 to 1045). The ends of the rivet shall be finished flush with the adjacent sides of the tool. The rivet shall be of proper size and shall be so fitted in the pliers that there will be no excessive movement or other indications of looseness between the tool halves and the rivet when in any operating position. The rivet shall possess a hardness of not less than 20 nor more than 50 on the Rockwell "C" scale when tested in accordance with 4.3.1.

3.9 Spiral mechanism. The spiral mechanism shall be capable of producing a close, uniform twist in safety wire without binding of or damage to the mechanism. This mechanism shall consist essentially of a spiral rod and housing, a spiral rod guide, and a pull knob. This mechanism shall be mounted between the handles of the tool and shall be permanently affixed to one handle by welding in such a position that the longitudinal axis of the spiral rod shall correspond to the longitudinal axis of the plier. The mounting method shall not fail during or after completion of the test of 4.3.4. The design of this mechanism shall be such that when the spiral rod is withdrawn from the housing, the entire tool will be caused to revolve about its longitudinal axis. The total travel of the spiral rod during each withdrawal shall cause the tool to make the following minimum number of complete revolutions:

(a) 9-inch size - 3 revolutions.

(b) 12-inch size - 4 revolutions.

3.9.1 Spiral rod and housing. The spiral rod employed shall be made of steel and shall have multiple right-hand flutes or grooves. A steel housing shall be provided which shall completely encase the spiral rod when in the retracted position in order to protect the rod from dirt or damage. A suitable stop shall be provided to limit the travel of the spiral rod and prevent accidental separation of the rod from the housing. Suitable means shall be provided to retain the spiral rod in the retracted position within the housing when the tool is held in a vertical position with the handle ends pointed down.

3.9.2 Pull knob. The spiral rod shall be equipped with a suitable metal pull knob for pulling the spiral rod out of the spiral rod housing, and pushing it back in, if it is not spring-loaded to retract automatically. That portion of the knob designed for gripping with the thumb and forefinger shall be not less than 11/16 inch in diameter and if cylindrical shall be knurled to prevent slipping from the fingers in operation. In the event the knob is flared to prevent slippage of the fingers, knurling will not be required. The design of the knob shall permit a comfortable, adequate grip by the operator. The pull knob shall be securely mounted to the spiral rod and shall be capable of passing the test in 4.3.4.

3.9.3 Spiral rod guide. A device shall be provided which will act as a guide for the spiral rod. The design shall be such that it will cause the wire twister to revolve when the spiral rod is withdrawn. If a sleeve type guide is used, it shall be of bronze, nylon, or other suitable bearing material. The bearing shall be capable of passing the test in 4.3.4. The bearing surface of the sleeve shall have a surface finish no rougher than 63 r.m.s. and shall have not less than 1-inch long bearing surface. If a caged ball bearing and pawl type guide is used, the device shall employ at least 2 hardened and ground steel balls. The balls shall be at least 1/8 inch in diameter. In either case, the design of the guide and the material used shall be such as to provide long life and ease of operation to the spiral mechanism.

3.9.4 Locking device. The wire twister shall be equipped with a slide action locking device sufficiently durable to withstand maximum usage. The sliding sleeve design of the locking device shall operate on a longitudinal axis with the handles and shall be positively secured to prevent disengagement from the locking device without interfering in its intended operational range of movement. There shall be a 4.0 to 1 minimum ratio between the distance from the end of the jaw to the center of the rivet and the center of the rivet to the point of locking facility on the 9-inch nominal size plier. There shall be a 4.7 to 1 minimum ratio between the distance from the end of the jaw to the center of the rivet and the center of the rivet to the point of locking facility on the 12-inch nominal size plier. The locking device shall be conveniently located and capable of being operated with one hand when gripping and releasing wire. The locking device shall be secured to the handle with a continuous weld. The device shall be so designed that when the plier jaws are closed, the handles must be further compressed in order to engage the lock pin.

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The resultant grip between the plier jaws in this locked position shall be such as to firmly and securely grip any size lockwire from 0.020 to 0.060 inch diameter. Release of the locking device shall be accomplished automatically by a spring-loaded device when the handles are compressed while holding the pliers in the hand. The locking device shall not interfere with the user's hands when performing lock wiring operations.

3.10 Wire twister pliers shall conform to table I for the size specified (see 6.1) and shall be similar to figure 1 and meet the dimensional requirements of figure 2.

TABLE I. Dimensions of pliers, wire twister, with side cutter

Nominal size	A ±1/2	B ±1/4	C
Inches	Inches	Inches	Inches
9	8-7/8	1-1/2	10-1/4 ±1/4
12	10-3/16	1-3/4	11-3/8 ±1/2

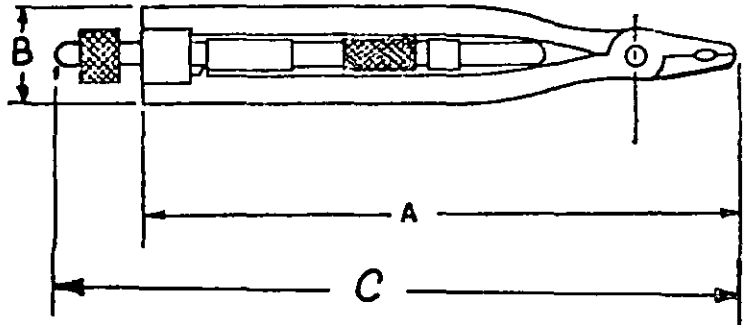


FIGURE 1. Pliers, wire twister, with side cutter.

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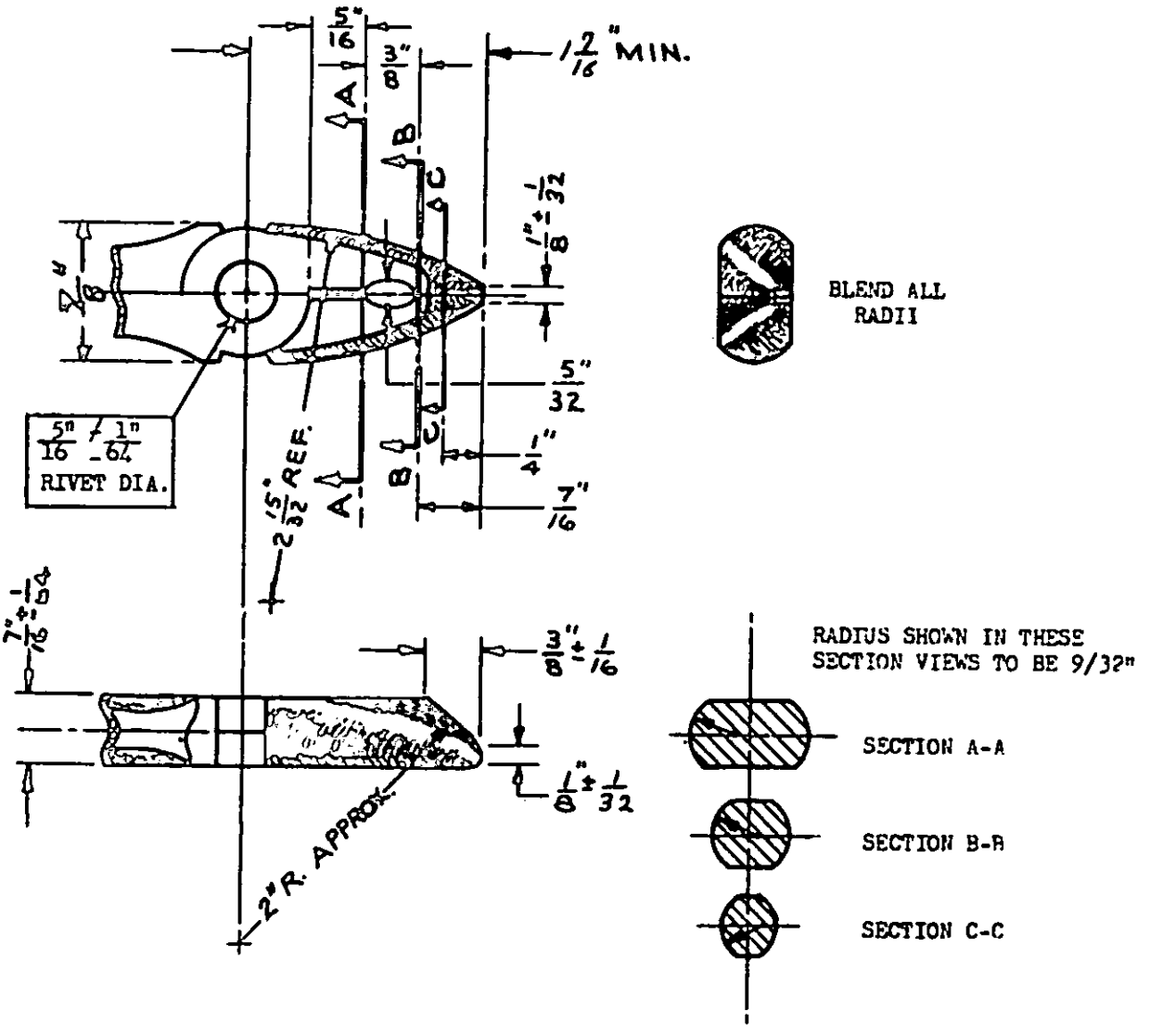


FIGURE 2. Nose design, pliers, wire twister.

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3.11 Workmanship. Workmanship shall conform to good commercial practice for the quality and grade of pliers covered by this specification. Surfaces shall be free of defects in surface finish (see 3.3) or coating (see 3.4) which would affect durability, serviceability, or appearance, or cause injury to the hands of the user of the tool. Forged parts shall be free of laps, cracks, and pits. Welded areas shall be neatly and smoothly blended with the adjoining members.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection of materials and components. In accordance with 4.1, the supplier is responsible for insuring that materials and components used were manufactured, tested, and examined in accordance with referenced subsidiary specifications and standards to the extent specified herein, or, if none, in accordance with this specification.

4.1.2 Sampling for lot acceptance inspection.

4.1.2.1 Lot. All pliers offered for delivery at one time shall be considered a lot for the purpose of acceptance, inspections, and tests.

4.1.2.2 Sampling for inspection. A random sample shall be selected from each lot in accordance with the provisions of MIL-STD-105 at inspection level II.

4.1.2.3 Sampling for tests. One sample plier shall be selected at random. If the sample fails to comply with any of the tests, two additional samples shall be selected. If any additional sample fails any of the tests, pliers represented by the samples shall be considered as not conforming to the requirements of this specification.

4.1.2.4 Lot acceptance inspection. Lot acceptance inspection shall be applied to each item in accordance with 4.1.2.2 and 4.1.2.3 prior to being offered for acceptance under the contract. Lot acceptance inspection shall consist of the following:

(a) Examination (see 4.2).
(b) Tests (see 4.3).
(c) Inspection of preparation for delivery (see 4.4).

4.2 Examination.

4.2.1 Visual and dimensional examination. The completely assembled end item shall be examined for defects in finish, construction, workmanship, marking, and nonconformance with dimensional requirements. Defects shall be classified in table II. The inspection level shall be level II with an Acceptable Quality Level (AQL) of 1.5 percent defective for major defects and 2.5 percent defective for minor defects.

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TABLE II. Classification of defects in accordance with MIL-STD-105

Categories	Defects
Critical:	
1	None defined.
Major:	
101	Size not as specified.
102	Material not as specified.
103	Excessive side play of the jaws or joints.
104	Jaws cannot be opened by the action of one hand.
105	Gripping surfaces defective or not properly scored.
106	Cutting edges defective, evidence of nicks, burrs, or "wire edges", not aligned, or not provided with adequate clearance or recess, or fail to make contact throughout the entire length when in the closed position.
107	Handles not smooth; evidence of fins, burrs, rough edges, or sharp corners.
108	Coating (if required) missing, incomplete, defective, or not as specified.
109	Maximum jaw opening at top of jaws not as specified.
110	Any failure to conform to specified requirements that may affect usability.
Minor:	
201	Jaw dimensions not within specified tolerances.
202	Overall length not within specified tolerances.
203	Distance between handles when in the closed position not within specified tolerances.
204	Jaw thickness and width not uniformly tapered from the joint.
205	Marking, manufacturer's name or trademark missing, illegible, incorrect, or not permanently marked.
206	Any failure to conform to specified requirements that is not likely to affect usability.

4.3 Tests and procedures. A sample of pliers selected in accordance with 4.1.2.3 from the random sample selected in accordance with 4.1.2.2 shall be tested as specified in 4.3.1 through 4.3.4.

4.3.1 Hardness. Pliers shall be tested for hardness in areas as shown in figure 3. The pliers shall be tested with the jaws open and properly prepared to insure accurate readings. Serration readings shall be made within 1/8 inch of the edge. Where three readings are taken, the average of these three readings shall be used.

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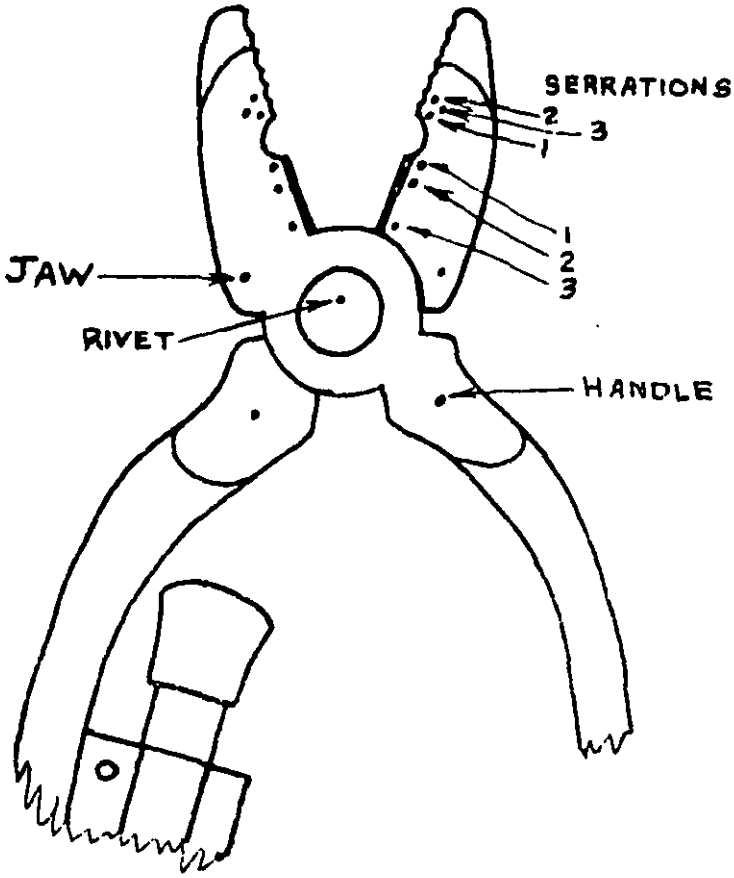


FIGURE 3. Typical locations for hardness readings.

4.3.2 Wire twisting and gripping tests. Two types of safety wire shall be used for the wire twisting tests: 0.020 inch diameter single strand annealed copper wire and 0.051 inch diameter annealed nickel-chromium-iron alloy (Inconel) safety wire. Sample wires shall be cut to 18 inch lengths, ±1/4 inch. Examination of pliers shall show no visible signs of cracks, undue wear, faulty operation, deformation, looseness, misalignment of plier parts, or any other harmful defect, after conducting the tests specified in 4.3.2.2 and 4.3.2.3. Handles shall be measured before and after gripping test, and shall not show more than ±1/16 inch permanent set.

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4.3.2.1 Test equipment. The equipment required for performing the wire twisting tests shall consist of two weights. Each weight shall have an eyebolt, "U" bolt, "J" bolt, or other suitable coupler securely attached to the weight so that a test wire may be looped through the coupler to lift the weight in an approximately balanced position. The eyebolt or other coupler shall have a 1/2 inch to 5/8 inch nominal cross section diameter at the point contacted by the test wire when it is used to lift the weight. The weight may be designed so that it can be prevented from rotating on its vertical axis during the wire twisting test, provided it is not restrained from being lifted. The size of the weight for performing the test specified in 4.3.2.2 shall be 35 lbs. ±1/2 lb. For the test specified in 4.3.2.3, the weight shall be 7 lbs. ±1/2 lb.

4.3.2.2 Test procedures. The 0.051 inch diameter Inconel wire shall be attached to the 35 lb. weight by passing one end through the eyebolt or other coupler being used, and then gripping both free ends of the wire with the jaws of the pliers. The ends of the wire shall be inserted side-by-side between the plier jaws to a depth of approximately 1/2 inch, and shall provide from 7-1/2 to 7-7/8 inches of free wire extending beyond the coupler for twisting. The jaws shall be locked in the gripping position by means of the locking mechanism. The wire shall then be twisted by gripping the spiral rod pull knob with the thumb and forefinger of one hand and pulling vertically against only the resistance of the weight until the spiral rod reaches its end of travel. The rod shall then be returned to its retracted position and the operation shall be repeated a sufficient number of times to produce a minimum of 10 uniform tight twist of 360° each in the wire. Without releasing the grip on the wire, the pliers shall then be made to lift the weight by means of the twisted wire, and shall hold the suspended weight for a minimum period of 15 seconds.

4.3.2.3 Small wire twisting test. The 0.020 inch diameter copper wire tests shall be conducted in the same manner as that specified in 4.3.2.2, except that the 7 lb. weight shall be substituted for the 35 lb. weight, and not less than 16 complete twists shall be made in the wire. In addition, after performing the lifting test, and before removing the twisted wire from the coupler, the grip of the plier jaws shall be released from the ends of the wire, transferred to, and locked on the twisted wire at a point approximately half-way between the free ends and the coupler. The grip shall then be released and transferred back to the ends of the wire, and the lifting test again performed. If the wire breaks at any point it was gripped with the plier jaws, the pliers shall be considered as having excessively weakened the wire.

4.3.3 Cutting tests. Each sample wire twister plier shall be subjected to the test of 4.3.3.1, following which the samples shall be subjected to the test of 4.3.3.2.

4.3.3.1 Wire cutting test. Each wire twister shall snap cut at the joint end of the cutting edges, three sample pieces of 0.041 inch diameter stainless steel wire conforming to QQ-W-423 composition 316, condition A, and three single pieces of 0.051 inch diameter Inconel wire as specified in 4.3.2. All cuts shall be clean in nature. The load required to completely sever the wire shall not exceed 300 inch-pounds. Jaw spreading exceeding the requirement of 3.7 shall be cause for rejection. If sample complies, it shall be subjected to the test of 4.3.3.2.

4.3.3.2 Bond paper cutting test. Paper for this test shall be either rag or chemical wood bond paper. The average thickness of the paper shall not exceed 0.004 inch. Paper shall be in strip form, the width of the strip measuring 1/16 inch less than the length of the cutting edge of the wire twister plier. The strip of paper shall be centered between the ends of the cutting edges when the cut is being made. The cutters shall completely and cleanly sever the paper when a load not to exceed 75 inch-pounds is applied.

4.3.4 Pull knob security test. With the spiral rod in the retracted position, a 100 lb. load shall be attached to the pull knob. A force shall then be applied to the pliers which will simultaneously extend the spiral rod and lift the 100 lb. load. The knob shall be capable of lifting and holding the 100 lb. load for one minute. After completion of the test, the knob shall not have slipped or loosened in any way. The guide bearing of 3.9.3 shall have remained in place with no visible deformation. In addition, the spiral rod shall function as prior to the test.

4.4 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the preservation, packaging, packing, and marking comply with the requirements of PPP-P-40.

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5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking shall be in accordance with PPP-P-40. Preservation, packaging, and packing shall be level A, B, or C, as specified (see 6.1).

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Size of plier required (see 1.2).
(c) Coating, if particular type is required (see 3.4).
(d) Applicable levels of preservation, packaging, and packing (see section 5).

MILITARY INTERESTS:

DOD has waived coordination on revisions and amendments to this Federal Specification until further notice.

Preparing activity:

GSA-FSS

Orders for this publication are to be placed with the General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 35 cents each.

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